

ARYENDRA PANDEY

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SKILLS

Programming: Python, C++, JavaScript, SQL
Backend & Software: FastAPI, REST APIs, Git, Linux, CMake
AI / ML: PyTorch, YOLOv8, OpenCV, ONNX, INT8 Quantization, ChromaDB, Streamlit
Web Technologies: Next.js, TypeScript, React, Tailwind CSS
Tools & Platforms: GitHub, CUDA, Splunk, Wireshark

WORK EXPERIENCE

Software Developer Intern

April 2026 – May 2026

Tridit Digital

- Developed and customized AEM content components, templates, and publishing workflows for an enterprise web platform.
- Implemented automated content tagging and personalization features within AEM pipelines, reducing manual editorial effort.

IT Infrastructure Intern

June 2026 – Present

XtraNet Technologies Limited, Bhopal

- Administered enterprise network infrastructure covering system configuration, fault diagnosis, and client-side issue resolution.
- Supported server environment operations, maintaining uptime and reliability of production systems.

PROJECTS

Brain Tumor Detection — Hardware-Aware Inference Optimization | [GitHub](#)

PyTorch · YOLOv8n · ONNX · CUDA · INT8 Quantization · LayerCAM · Streamlit

- Trained YOLOv8n on 7,200 MRI scans across 4 tumor classes, achieving **89.18% mAP@0.5** with CUDA-accelerated inference at **6.23 ms/image** on an RTX 3050 GPU.
- Applied FP32 → INT8 static quantization, reducing model size by **47.89%** (5.94 MB → 3.09 MB) with only 0.48% mAP drop, confirming edge deployment viability.
- Deployed a Streamlit diagnostic dashboard with LayerCAM explainability, IoU scoring, and FP32 vs INT8 inference benchmarks.

NeuroSentry — Log-Based Threat Detection System | [GitHub](#)

Python · FastAPI · Scikit-learn · Log Parsing · React · Vite

- Built a modular threat detection pipeline: parsed system/audit logs, engineered behavioral features, and applied an ML classifier for near real-time anomaly detection.
- Developed a FastAPI backend with dedicated REST endpoints for log ingestion, feature extraction, and model inference; served predictions to a React + Vite dashboard displaying live alerts and event statistics.

SoTrail — AI Travel Planning Platform | [GitHub](#)

FastAPI · ChromaDB · RAG · Llama · Amadeus API · Next.js · TypeScript

- Engineered a FastAPI backend with a retrieval-augmented generation (RAG) layer using ChromaDB for vector search and a Llama-based chatbot for personalized travel recommendations.
- Integrated Amadeus API for live flight and hotel data; built a Next.js + TypeScript frontend with itinerary planner, chatbot panel, and trip analytics.

QoS Degradation Predictor — Network Systems ML | [GitHub](#)

C++17 · CMake · Logistic Regression · Network Simulation

- Implemented a time-driven packet-switched network simulator in C++17 with multi-flow FIFO queues; trained a logistic regression model on simulated traffic metrics for QoS degradation detection.

EDUCATION

Manipal Institute of Technology, MAHE

2023 – Present

B.Tech — Electronics and Communication Engineering

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming, Machine Learning

CERTIFICATIONS

[Google Cybersecurity](#)

[AWS Cloud Practitioner](#)

[Meta Front-End Dev](#)

[Meta JavaScript](#)

EXTRACURRICULAR ACTIVITIES

- **Core Committee Member** — TechTatva 2025, Technical Fest, MIT Manipal
- **Organising Committee Member** — Revels 2025, Cultural Fest, MIT Manipal
- **Management Committee Member** — Aaina Dramatics, MIT Manipal